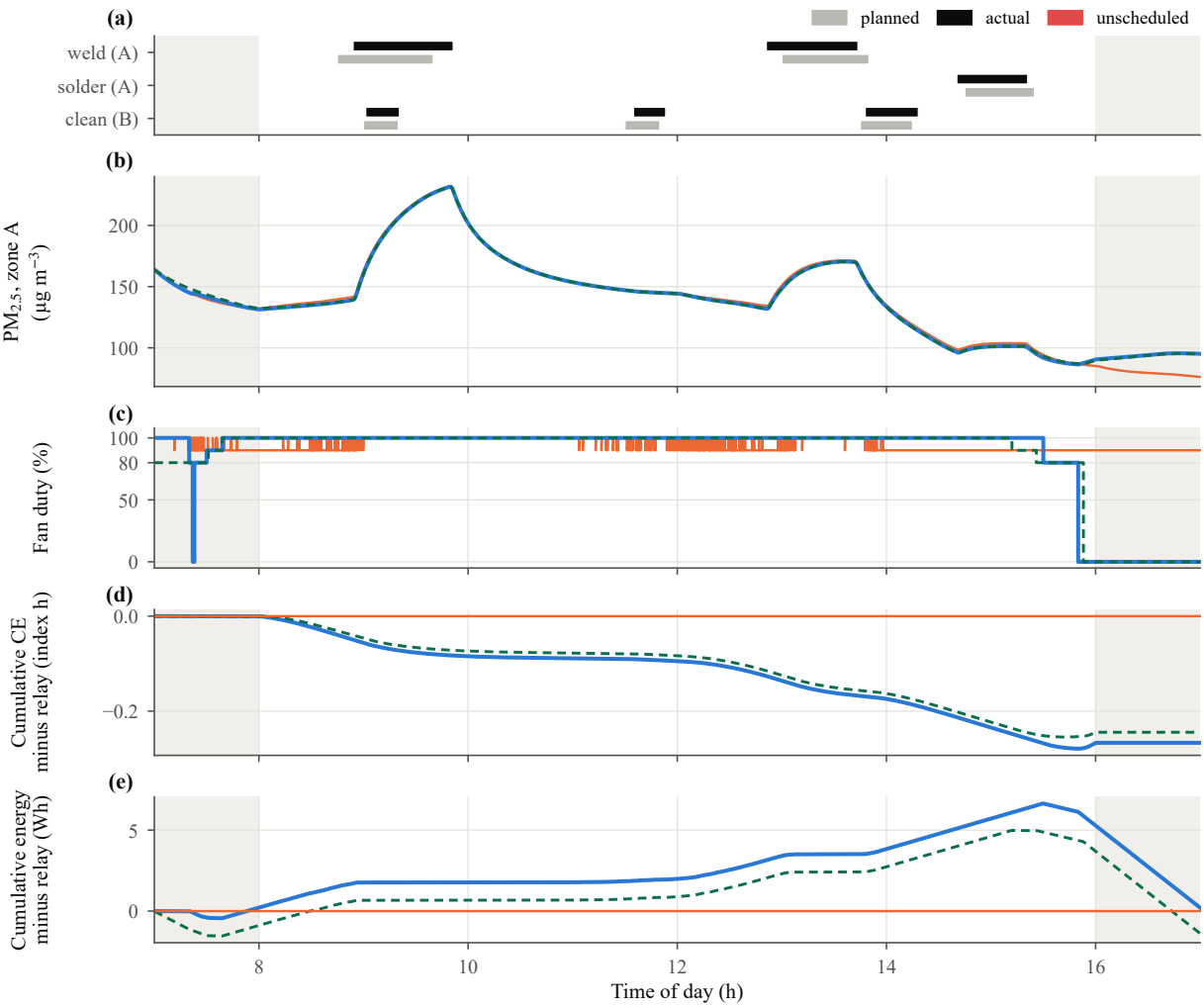

(a)
planned
actual
unscheduled
weld (A)
solder (A)
clean (B)
(b)
PM2.5, zone A (µg m−3)
200
150
100
(c)
Fan duty (%)
100
80
50
0
(d)
Cumulative CE minus relay (index h)
0.0
−0.2
(e)
Cumulative energy minus relay (Wh)
5
0
8
10
12
14
16
Time of day (h)
Relay (baseline)
EMPC, schedule feedforward
Rollout costate law (CL-r)